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Effectiveness of EMI in higher education: A multi-level meta-analysis

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Abstract

This study is one of the first comprehensive multi-level meta-analyses investigating the effects of English Medium Instruction in higher education. It synthesizes a total of 41 samples, comprising a total of 8,747 participants, from 28 quantitative studies, including 23 samples on content learning (7,659 participants) and 18 on English performance (1,088 participants). A comparison of EMI students' post-test and pre-test scores (within-group comparison) revealed that students significantly improved on measures of content knowledge (d = 1.57, p < .001) and English proficiency (d = 0.81, p < .001), although we acknowledge that this finding does not account for external factors or establish causality. A comparison of post-test performance between EMI and non-EMI groups (between-group comparison), which aimed to assess EMI's effectiveness relative to non-EMI, showed that EMI students achieved comparable outcomes in content learning (d = 0.13, p = 0.14) and significantly outperformed non-EMI students in English development (d = 0.33, p = 0.009). Moderator analyses further revealed that studies with methodological limitations, such as selection biases for within-group comparisons or unaddressed group homogeneity for between-group comparisons, often overestimate the effectiveness of EMI for content learning. Based on our findings, we offer practical implications for the implementation of EMI in higher education.

Keywords: Content learning; English medium instruction; English learning; Multi-level meta-analysis

The considerable body of research that exists on the phenomenon and practice of English Medium Instruction (EMI) now includes a substantial and growing interest in the success or 'effectiveness' of the practice of teaching academic subjects through the medium of English to students whose first language (L1) is not English (e.g., Aguilar & Muñoz, 2014; Aizawa et al., 2024; García-Álvarez de Perea & Ramírez-García, 2024; Guo et al., 2018; Lei & Hu, 2014; Rogier, 2012). However, we should note that interest in 'effectiveness' (the term used in this paper) has not overshadowed the continued concern that some authors have expressed concerning the possible additional (potentially negative) impact that EMI might have on a given language, on national identity, or on a socio-cultural group. It is, therefore, of some importance to attempt to define effectiveness and subsequently evaluate evidence of whether EMI has been effective to an extent which counterbalances arguments related to its potential negative impacts.

Effectiveness in EMI higher education contexts (the focus of our study) is generally evaluated, or specifically measured, according to two dimensions: gains in academic content knowledge, and gains in English language proficiency (Peng & Xie, 2021; Rose et al., 2019). In the case of the first dimension, stakeholders would want to ascertain, through research, that content outcomes were at least as good as outcomes achieved through L1 medium instruction (Macaro, 2022). However, given that, for many disciplines in higher education, very large quantities of research are published in English, then it would not be unreasonable to expect that EMI might, in the long run, outperform its non-EMI counterpart. In the case of the second dimension – increase in English language proficiency – again, it is not unreasonable to expect that students would improve their English more quickly and more deeply in EMI programmes than in non-EMI equivalents. Indeed, the expectation of greater English proficiency is a cornerstone of a number of national language policies in which there is an ambition to develop much higher levels of bilingualism among the population (see, for example, Ferrer & Lin, 2024 regarding Taiwan; Rose & McKinley, 2018 regarding Japan; and Fang, 2018 regarding China).

To evaluate the effectiveness of these two dimensions, however, is not as simple as it sounds. First, in terms of content learning, there are a number of sub-factors that need to be taken into account: should researchers take into account prior content learning at high school (or in a previous phase of education) and in which language that took place? are there disciplinary differences that might allow faster learning through EMI in one discipline as opposed to another? Do the various and differing models of EMI, such as full and partial programmes (see below), make a difference as to the likelihood of recognizable content learning effectiveness? In terms of English proficiency effectiveness, there are also obvious factors to be considered: what aspect of the English language is being measured (vocabulary; grammatical correctness;

fluency; reported self-confidence)? are these different aspects likely to bias one group being measured against another? how does one control for lack of similarities between groups (e.g., social status where one group may have received intensive private tuition)?

So, attempting to answer the question as to the effectiveness of EMI is complex, but its complexity should certainly not deter researchers from attempting to do so, rather, it should encourage them to exercise caution in summarizing their findings. The fundamental aim of this meta-analysis, therefore, is to add to existing research by synthesizing quantitative findings on the effectiveness of EMI in terms of content learning and English proficiency enhancement. While this meta-analysis was constrained by the limited number of available studies and could therefore examine only a small set of factors, we trust that readers will interpret our findings within the broader context of EMI research.

1. Literature review

1.1. Theoretical foundations of English Medium Instruction in higher education

As EMI has emerged as an educational phenomenon, there have been ongoing debates over where the research field should theoretically be positioned (Macaro & Aizawa, 2024). In a recent review of research, some scholars have noted: “While it may not be accurate to say there is a complete “lack” of theory in English Medium Instruction (EMI) research, it’s fair to argue that there is a need for more comprehensive, robust, and specific theoretical frameworks to guide this field of study” (Curle et al., 2024, p. 4). Nonetheless, there are certain cognitive and linguistic theories which can be applied to interpret content and language gains in an EMI educational environment.

1.1.1. Arguments for EMI

Relevant theories related to language and content development in EMI center around the perceived benefits of greater levels of second language (L2) input and interaction afforded by learning content through English as an additional or second language. As Costa and Mariotti (2023) claim, “field exposure to input is the prime mover of the teaching-learning process and … its quality is fundamental for the development of discipline-specific knowledge with particular reference to university settings” (p. i). Macaro (2018) has previously noted that university EMI lectures are heavily characterized by large quantities of teacher input, further highlighting the applicability of input theories. Thus, it is this input, alongside academic readings and other teaching materials, that is thought to primarily drive learning processes in EMI (Costa & Mariotti, 2023). According to VanPatten (2009), input and interaction theories are embedded in the notion that when learner attention is simultaneously focused on meaning

and the properties of language, learning occurs most effectively. Parallels can also be drawn with theories of bilingual education, where the *Interdependent Hypothesis* purports that cognitive and linguistic capacity in one's L1 enhances the development of capacities in the L2 in such a way that cognitive or literacy skills develop more rapidly in the L2, if already acquired in the first (Cummins, 1979). Thus, according to these theories of SLA and bilingual education, EMI in later stages of education should, in theory, lead to positive outcomes in language development *and* drive content learning.

1.1.2. Arguments against EMI

There exist other educational theories that suggest a possible negative effect of EMI on learning, one of which is *Cognitive Load Theory*. Cognitive Load Theory posits that learning processes may be limited by the cognitive load, that is, the amount of information one's working memory can process at any given time (Sweller, 2011). Drawing upon this theory, Curle et al. (2024) illustrate:

> "For instance, a student who is not proficient in English might experience a high extraneous cognitive load when trying to understand a lecture in English, leaving less cognitive resources for processing the actual academic content of the lecture. According to the Cognitive Load Theory, this could potentially lead to poorer academic outcomes for the student." (p. 4).

Similarly, SLA theories of *comprehensible input* have seen a resurgence in research applicability due to their relevance to EMI (Costa & Mariotti, 2023; Macaro, 2018). The theory of comprehensible input (e.g., Krashen, 1982) posits that the level of language must be within the realms of understandability for students, or just above their current level of proficiency, for learning to occur. Similar notions can be found in the *Threshold Hypothesis* (Cummins, 2021), which proposes that learners require a minimum threshold level of proficiency in an L2 before they can benefit from its use as a medium of instruction in school, otherwise there may be negative effects on learning.

Taken together, these linguistic and cognitive theories suggest that while EMI affords opportunities for greater content and language input for many learners, if lecture and reading content is too difficult or beyond the linguistic capabilities of students, learning outcomes may be negatively affected. Understanding how EMI affects learning outcomes, therefore, is a matter of both practical and theoretical importance to the field.

1.2. Synthetic evidence on the effectiveness of English Medium Instruction

In this section, we present previous efforts to synthesise empirical research on the effectiveness of EMI and their implications and limitations; the foundation upon which, the present meta-analysis was designed and conducted. While medium of instruction has been investigated in several previous research syntheses of Content and Language Integrated Learning (CLIL) (see, for example, Kaiypova et al., 2025; J.H. Lee et al., 2023), there have been fewer efforts that have explicitly and exclusively targeted EMI research.

1.2.1. Synthetic evidence on the effectiveness of EMI for content learning

In one of the first full-scale systematic reviews on this topic, Macaro et al. (2018) conducted an in-depth review of 83 studies on EMI conducted in higher education and identified several themes surrounding these studies, two of which are relevant to the topic of this meta-analysis. For one of these themes–the impact of EMI on content comprehension and learning, only four studies were identified, and these studies showed some contrasting results. Overall, Macaro et al. suggested that due to the small number of studies and their methodological shortcomings (e.g., validity of instruments, the issue of homogeneity of EMI and non-EMI groups), they could not draw any meaningful conclusions about the effects of EMI. Graham et al.'s (2018) systematic review focused on EMI and CLIL studies that had measured student learning outcomes. Of the 25 studies identified in their review, only three were studies that examined content learning outcomes in higher education, and these showed inconsistent results, with two studies finding no significant difference and one study showing the superiority of the EMI condition over its non-EMI counterpart. Finally, Peng and Xie's (2021) meta-analysis focused exclusively on Chinese studies in higher education and compared the effectiveness of the EMI condition with that of the Chinese-medium instruction (CMI) condition. Regarding content learning outcomes, their results revealed that the EMI group performed significantly better than their CMI counterparts, with this difference showing an overall medium effect size (d = 0.67, SE = 0.134, $p < .001$). Commenting on this rather unexpected finding, they explained that approximately 74 % of their dataset was from medical disciplines, which may have influenced the overall effect size; indeed, no significant difference was found when only samples related to non-medical disciplines were considered ($d = -0.06$, $p > .05$, 95 % CI [–0.18, 0.05]).

1.2.2. Synthetic evidence on the effectiveness of EMI for English Learning

The aforementioned systematic review by Macaro et al. (2018) identified 10 studies on English learning outcomes and found that they generally showed improvement, but only in specific skills or subcomponents of skills. Macaro et al. further highlighted the need to distinguish between general and academic English proficiency when measuring the

effectiveness of EMI, as it is expected that EMI may be more beneficial for the latter, suggesting that the type of English proficiency may be a potential moderator of the effectiveness of EMI. In the systematic review by Graham et al. (2018), only one study (Yang, 2015) was identified regarding language learning outcomes, and its results showed that the EMI (CLIL) condition favored receptive skills, though this study had some methodological issues (e.g., small sample size and no comparison group involved in the study). The meta-analysis by Peng and Xie (2021) additionally analyzed English learning achievements with 12 independent samples, and showed that the EMI group had better English learning outcomes, with a large effect size (d = 1.58, SE = 0.34, p < .001).

Although the systematic reviews by Macaro et al. (2018) and Graham et al. (2018) provide a detailed synthesis of existing studies on the impact of EMI on content and English learning outcomes, the results of these reviews are far from conclusive on the effectiveness of EMI on student learning outcomes in higher education. It is also noteworthy that more studies have been conducted on this topic since these research syntheses were conducted, pointing to the need for an updated meta-analysis that includes more recent studies. Furthermore, given Peng and Xie's (2021) meta-analysis focused exclusively on the Chinese context, a more comprehensive meta-analytic effort that includes other educational contexts seems necessary– a gap that the present meta-analysis aims to fill.

1.3. Key features influencing the effectiveness of English Medium Instruction

According to a recent meta-analytic structural equation modelling study on the mechanisms underlying successful EMI programmes in higher education (H. Lee et al., 2025) which analyzed 50 studies, comprising a total of 15,032 participants, three key features were identified as influencing EMI success. These features include L1-English proximity (i.e., whether the L1 and English are linguistically related), the EMI context (i.e., full EMI vs. partial EMI), and institutional support (i.e., whether any institutional support was provided or not). These findings were derived from a comprehensive review of theoretical frameworks and existing literature, with success often measured by outcomes, such as students' GPAs or test results following EMI programmes.

Another meta-analysis by J.H. Lee et al. (2025) focused on the effectiveness of EMI at the primary level, rather than higher education. Their study, based on 28 samples (based on a total of 214,103 participants), suggested that learners' age, EMI context, and confirmation of homogeneity were potential moderators influencing both content and language learning. In secondary education, studies by Kaiypova et al. (2025) and J.H. Lee et al. (2023), which analyzed 29 samples (36,905 participants) and 44 samples (7434 participants), respectively,

identified the following potential moderators: L1-English proximity, EMI context, homogeneity confirmation, test language, the nature of the content subject, and the target linguistic dimension.

When critically reviewing these findings to assess their relevance to EMI in higher education, across diverse learner backgrounds, instructional contexts, and methodological considerations, eight key features emerge as influential to EMI effectiveness. Regarding learner backgrounds, learners' L1 and its proximity with English is expected to play a significant role, as learners whose L1 shares linguistic or structural similarities with English may adapt more easily to EMI compared to those whose L1 is unrelated.

Next, we identified factors related to instructional contexts. First, institutional support, such as the provision of language workshops, tutoring, and faculty training, would be a pivotal factor in facilitating EMI success. Second, the extent of EMI exposure, whether a programme adopts full or partial EMI, can affect both content knowledge acquisition and English language development. Third, the duration of EMI programmes, as the outcomes of short-term (semester-long) programmes may differ from those of longer-term programmes spanning one to four years.

The final group of factors concern methodological aspects of research. First, the subject matter taught through EMI is identified as a potential moderating factor. For instance, STEM courses may pose fewer linguistic challenges for students due to the routinized and domain-specific nature of their terminology, which tends to be more predictable and less discursive than in non-STEM disciplines (see e.g., Airey, 2012; Pecorari & Malmström, 2018). Second, the measurement of English learning outcomes has been found to vary, with some studies focusing on academic English skills tailored to higher education, while others measure general English proficiency, which may not fully capture the language demands of EMI programmes. Third, homogeneity confirmation has been pointed out as a key factor influencing the reliability of EMI research findings, as studies that failed to check homogeneity in their samples would tend to include more motivated and talented participants. Similarly, entry prerequisites—such as requiring specific levels of English proficiency before enrolment—can serve as gatekeeping mechanisms to ensure that students are adequately prepared and that samples are more comparable.

These eight features collectively underscore the multifaceted nature of EMI in higher education and the variability in its effectiveness across institutions and contexts.

2. Current study

In this study, we conducted a meta-analysis to systematically synthesise prior empirical research on the effectiveness of EMI programmes in higher education. We adopted a multi-level meta-analytic approach to account for multiple effect sizes within studies reporting more than one measurement. This method facilitated the calculation of overall average effect sizes and the conduct of subgroup analyses to examine moderator effects while addressing the multi-level structure of the dataset (e.g., H. Lee et al., 2019; J.H. Lee et al., 2023). Specifically, effect sizes at the measurement level were nested within the sample level, enabling more precise estimates and mitigating dependence issues arising from multiple effect sizes within a single sample (Hox et al., 2017). Through this approach, we aimed to investigate the effectiveness of EMI programmes for enhancing learner's content knowledge and English proficiency. This was examined from both non-experimental perspectives (e.g., pre-test and post-test measurements within EMI groups) and (quasi-) experimental perspectives (e.g., post-test comparisons between EMI and non-EMI groups).

3. Research questions

In total, we formulated three research questions to achieve our research aims. RQ1 and RQ2 sought to estimate the overall average effect size of EMI programmes in higher education for content learning and English learning, respectively. These questions address an insufficiently explored issue: the effectiveness of EMI programmes from a cost-benefit perspective (Macaro, 2022). RQ3 explores features identified in existing literature as possible moderators of EMI effectiveness, and which of them significantly impact the effectiveness of EMI programmes.

The following are the three research questions and the expected findings.

3.1. Research question 1: What is the overall average effect size of EMI programmes in higher education for content learning?

RQ 1.1: What is the overall average effect size for content learning within-group samples (from start to end of the programme)?

RQ 1.2: What is the overall average effect size for content learning between-group samples (EMI vs. non-EMI)?

Expected findings: EMI programmes will be beneficial to students in terms of improving their content knowledge (per within-group samples). However, per between-group samples, given the proposition of cognitive load theory, they will be less effective in terms of content learning than non-EMI programmes (in which instruction is given in students' L1).

3.2. Research question 2: What is the overall average effect size of EMI programmes in higher education for English learning?

RQ 2.1: What is the overall average effect size for English learning within-group samples (from start to end of the programme)?

RQ 2.2: What is the overall average effect size for English learning between-group samples (EMI vs. non-EMI)?

Expected findings: Given ample opportunities for English input and interaction, EMI programmes will be beneficial to students in terms of improving their English proficiency (per within-group samples). Also, they will be more beneficial than their non-EMI counterparts (per between-group samples).

3.3. Research question 3: What are the key features that influence the impact of EMI programmes in higher education? And how do the identified features influence the impact of EMI programmes in higher education?

RQ 3.1: How do these features impact content learning in within-group and between-group samples?

RQ 3.2: How do these features impact English learning in within-group and between-group samples?

Expected findings: The identified features may influence the impact of EMI programmes in complex ways. Among them, homogeneity confirmation is expected to be the most significant. Particularly, studies that do not confirm the homogeneity of EMI and non-EMI groups are likely to report more positive findings in favor of EMI programmes, possibly due to selection bias—i.e., EMI groups may consist of more motivated or better-prepared students at baseline.

4. Method

4.1. Literature search

To construct the dataset for our meta-analysis, the first and fourth authors collaboratively searched the literature for empirical studies on the effectiveness of EMI in higher education. First, we conducted keyword searches for relevant databases, such as ERIC (EBSCO), ProQuest Social Science Premium Collection, Scopus, and Web of Science Core Collection using the following keyword combination of: [EMI OR CLIL] AND [English] AND [“higher education”] AND [randomized OR “control group” OR post-test OR quantitative OR cross-section OR matched OR “more effective” OR non-EMI OR non-CLIL OR “statistical difference” OR experimental].

The keyword combination for the meta-analysis was applied specifically to the abstract field of each study to ensure a precise and relevant selection of research. The first element, [EMI OR CLIL], identifies studies focused on EMI or CLIL. The second element, [English], confirms that the target language for EMI or CLIL is English. The third element, ["higher education"], ensures the studies are situated within higher education contexts. Finally, the fourth element, [randomized, "control group," post-test, quantitative, cross-section, matched, "more effective," non-EMI, non-CLIL, "statistical difference," experimental], filters for quantitative studies using experimental, comparative, or statistical methods.

Second, in addition to the database search, we conducted a backward search by reviewing the reference lists of systematic reviews and meta-analyses on this topic to identify additional empirical studies. Additionally, we carried out a forward search using Google Scholar's "cited by" function to locate empirical studies that cited these systematic reviews and meta-analyses after their publication.

Third, we compiled the studies identified through the database, forward, and backward searches, removed duplicates, and conducted a preliminary screening by reviewing their titles and abstracts to exclude irrelevant studies before proceeding to a full review.

Lastly, we performed a comprehensive review of the remaining studies to assess their eligibility based on the following four inclusion criteria: (1) the study must be written in English, (2) it must focus on EMI contexts in higher education, (3) it must aim to evaluate the effectiveness of EMI programmes for content learning, English learning, or both, and (4) it must report statistical data necessary for calculating an effect size (i.e., Cohen's *d*) for the effectiveness of EMI programmes, such as comparing EMI students' pre-test and post-test results (within-group comparison), comparing post-test performance between EMI and non-EMI groups (between-group comparison), or both. When a study did not fully provide statistical data, we contacted its authors to request the missing data.

4.2. Dataset construction, calculation of effect sizes and coding of moderator variables

After finalising the list of empirical studies to be included in the meta-analysis, the first and fourth authors collaboratively reviewed whether a study included more than one independent sample. If so, each sample was treated as a separate study.

We then calculated effect sizes in the form of Cohen's *d*, by extracting the relevant statistical data for each sample. If a sample had multiple measurements, we computed separate effect sizes for each. In traditional meta-analysis, we compute one effect size per sample to avoid the independent issue. However, our decision to use a multi-level meta-analysis allows us to address this challenge (see Data Analysis for details). For studies providing descriptive

statistics, such as means, standard deviations, and sample sizes, we used these values to calculate effect sizes. When inferential statistics were provided, such as the results of ANCOVA with pre-test scores as covariates or regression coefficients controlling for other factors, we used these estimates to calculate more precise effect sizes. For these calculations, we used a web-based effect-size calculator (Wilson, 2023).

After calculating the effect sizes, the first and fourth authors collaboratively coded moderator variables to assess whether each sample provided sufficient information on key features influencing the effectiveness of EMI. Although reporting inter-rater reliability is encouraged to enhance methodological rigor, we did not formally quantify discrepancies, as coding was conducted jointly and all differences were resolved through in-person discussion. (1) L1-English proximity was coded based on whether the learner's L1 and English are linguistically related, following the rationale proposed by H. Lee et al. (2022) and H. Lee and J.H. Lee (2024), using a web-based tool for quantifying genetic proximity (Beaufils, 2024). (2) Institutional support was coded based on whether language workshops, tutoring, and/or faculty training were provided. (3) EMI context was coded to distinguish between full EMI programmes, where all courses at the institution are taught in English, and partial EMI, where only selected courses are delivered in English. (4) Duration of EMI programmes was coded according to whether post-tests were conducted after short-term programmes (semester-long) or longer programmes spanning one to four years. (5) Subject matter was coded to identify whether the programme focused on STEM or non-STEM disciplines. (6) Measurement of English learning outcomes was coded based on whether the study focused on academic English skills or general English proficiency. (7) Homogeneity confirmation was coded based on whether within-group samples controlled for selection bias in EMI participants or whether between-group samples confirmed homogeneity between EMI and non-EMI groups. (8) Entry prerequisites were coded based on whether any prerequisite courses or specific levels of English proficiency were required for enrolment.

4.3. Data analysis: computation of overall average effect size and moderator analysis

Using the calculated effect sizes and coded moderator variables, we computed the overall average effect sizes of EMI programmes for content learning and English learning in both within-group and between-group samples. We then conducted moderator analyses to assess the extent to which the identified features influenced the effectiveness of EMI programmes for content learning and English learning in both within-group and between-group samples. We used STATA version 16 as the main statistical software for these data analyses. For interpreting the magnitude of Cohen's *d*, we adhere to the widely recognized standards of 0.2, 0.5, and 0.8, representing small, medium, and large effect sizes, respectively.

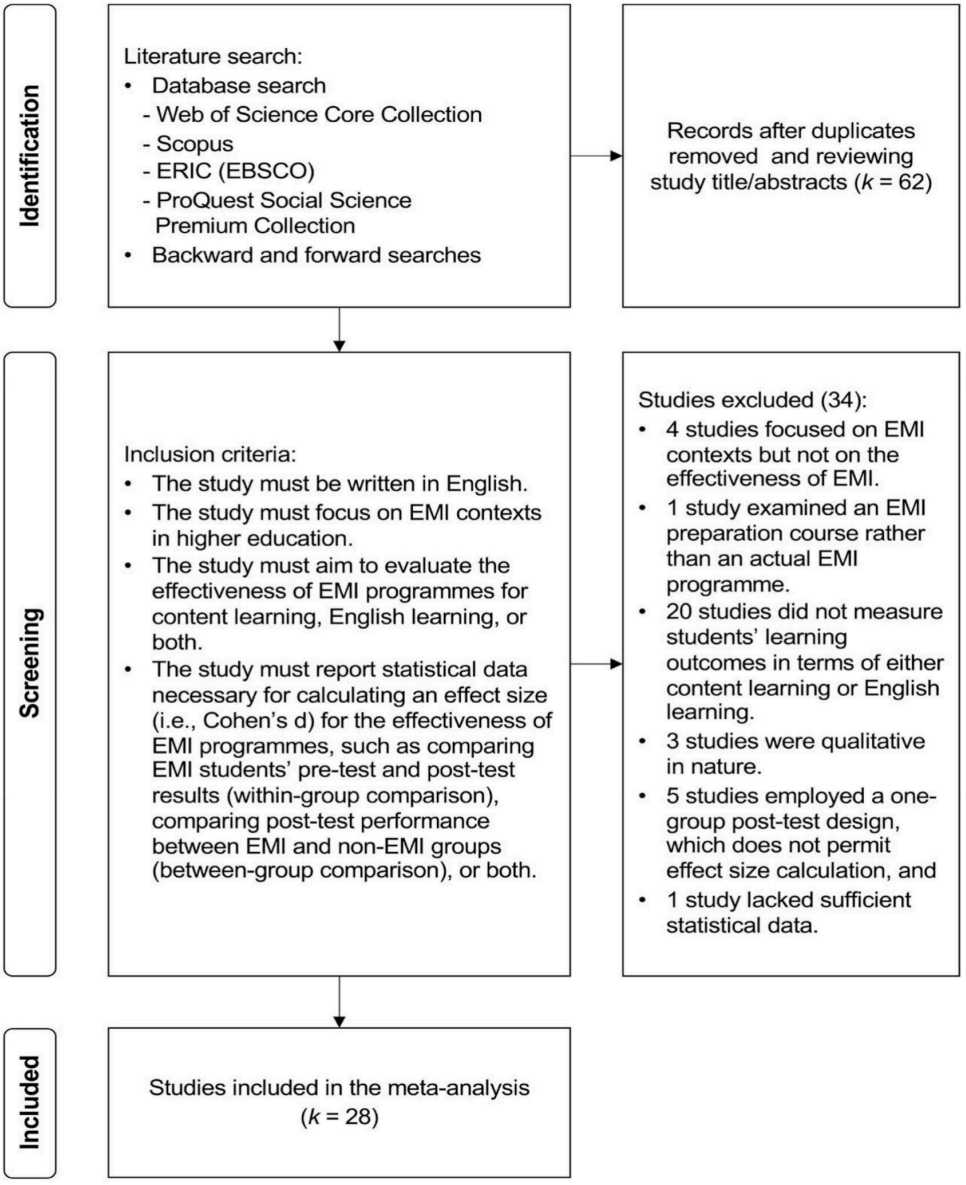


Fig. 1. Prisma flowchart for the literature search procedure.

First, to calculate the overall average effect sizes, we conducted a multi-level meta-regression analysis, treating the computed effect sizes as the dependent variable and using their associated standard errors to define the Level 1 variance, which reflects the sampling variability of individual effect sizes within studies. In this unconditional model (with no predictors), the intercept represents the overall mean effect size. This approach differs slightly from conventional multilevel regression because the Level 1 variance is not estimated from the data, but is treated as known (based on the standard errors of the effect sizes) hence, the term "variance-known" model (Hox et al., 2017). This specification allows us to model the nested structure of the dataset, with effect sizes clustered within samples, while assigning greater weight to more precise estimates (i.e., those from larger samples or with smaller standard errors).

When computing the overall average effect sizes, we included forest plots to provide visual cues for the reader's intuitive understanding of the distribution, magnitude, and variability of the effect sizes, as well as potential patterns in the data. Additionally, to assess potential biases, we evaluated small-study effects—commonly referred to as publication bias in meta-analyses—using funnel plots and the regression-based Egger's test in STATA software. We also performed outlier and influence diagnostics using the *metafor* package in R, which is instrumental in identifying and visualizing outliers through mathematical calculations. Our analysis revealed that none of the four groups—content learning (within-group and between-group) and English learning (within-group and between-group)—exhibited significant signs of publication bias when both criteria were considered simultaneously. While some groups showed significant signs of bias according to one criterion, this was not consistently supported by the other. Due to space limitations, the results and interpretation of the publication bias analysis are included in Appendix 1.

Second, to conduct moderator analyses, we extended the multi-level regression model by including moderator variables as independent variables in the equation. In this setup, the computed effect sizes remained the dependent variable, and the standard errors of the effect size estimates continued to serve as the Level 1 variance. Moderator variables, representing key features of EMI across diverse learner backgrounds, instructional contexts, and methodological considerations, were added to the model to examine their influence on the effectiveness of EMI. In this way, the analysis identifies which factors significantly influence the effectiveness of EMI programmes and quantifies the extent of their impact. This approach also allows us to explain some of the Level 2 variance (between-study variability) by linking it to specific key features, providing deeper insights into the conditions under which EMI is most effective. In this study, although we initially had 41 samples, comprising a total of 8747 participants, dividing them into four groups: content learning (within-group, between-group), and English learning (within-group, between-group), resulted in insufficient sample sizes for conducting multiple regression in the moderator analysis. As a result, we conducted moderator analyses using subgroup analysis, dividing samples by categorical moderator values. This approach yields less nuanced findings compared to multiple regression (meta-regression), which can isolate moderation effects while controlling for other variables, offering more precise and reliable estimates. The complete dataset at the sample level including study title, calculated effect sizes, and coded values for moderator variables, are included in Appendix 2.

5. Results

5.1. Literature search

Based on our database searches, we identified 60 studies from the *Web of Science Core Collection*, 6 from *Scopus*, 20 from *ERIC (EBSCO)*, and 20 from the *ProQuest Social Science Premium Collection*. An additional 26 studies were identified through backward and forward searches. After removing duplicates and irrelevant studies during the title and abstract screening, 62 studies remained for a full-text review. Applying our four inclusion criteria, we excluded 34 studies for the following reasons: (1) 4 studies focused on EMI contexts but not on the effectiveness of EMI, (2) 1 study examined an EMI preparation course rather than an actual EMI programme, (3) 20 studies did not measure students' learning outcomes in terms of either content learning or English learning, (4) 3 studies were qualitative in nature, (5) 5 studies employed a one-group post-test design, which does not permit effect size calculation, and (6) 1 study lacked sufficient statistical data. For the final case, we contacted the authors, who confirmed that the relevant data was unavailable. Consequently, 28 studies were included in the final dataset. The complete list of studies is in Appendix 3. Fig. 1 illustrates the literature search procedure using a PRISMA flowchart.

5.2. Dataset construction

Among the 28 studies (see Appendix 3 for the full list of references for the included studies), we identified a total of 41 independent samples, as 7 studies included multiple samples (20 in total), while the remaining 21 studies contributed one sample each. First, Aizawa et al. (2023) employed a pre-post design with a control group, featuring a between-group comparison of EMI and non-EMI groups and an additional within-group EMI analysis, contributing 2 independent samples. Second, Satayev et al. (2022b) included two EMI groups that measured both content learning and English learning. As these outcomes were treated separately, the study provided 4 independent samples. Third, Yang et al. (2019) conducted cohort comparisons between EMI and non-EMI groups for the academic years 2015 and 2016, resulting in 2 independent samples. Fourth, del Campo et al. (2016) offered cohort comparisons for EMI and non-EMI groups across three academic years (2009–2010, 2010–2011, and 2011–2012), yielding 3 independent samples. Fifth, Thi et al. (2023) examined EMI and non-EMI groups for both content learning and English learning, contributing 2 independent samples. Sixth, Guo et al. (2018) employed a pre-post design with a control group to assess both content learning and English learning. So, the study included two between-group comparisons of EMI and non-EMI groups for content and English learning and had two within-group comparisons for the EMI group across pre- and post-tests for content and

English learning. As content learning and English learning were treated as separate outcomes, the study contributed a total of 4 independent samples. Lastly, Salamanca and Montoya (2018) used a pre-post design with a control group, incorporating a between-group comparison of EMI and non-EMI groups and an additional within-group EMI analysis, contributing 2 independent samples. The details of the number of samples in each study and how they were coded are described in Table 1.

5.3. Calculation of effect sizes

For the 41 identified samples, we calculated effect sizes (i.e., Cohen's *d*), resulting in a total of 51 effect sizes. Thirty-three samples contributed one effect size each, while eight samples had multiple effect sizes, yielding 18 additional effect sizes in total. Specifically, the following studies had multiple measurements: Lin and Lei (2021) measured three outcomes (i.e., assignment, participation, and final exam for content learning); Satayev et al. (2022a) measured two outcomes (i.e., listening and grammar for English learning); Aguilar and Muñoz (2014) assessed two outcomes (i.e., listening and reading for English learning); Yang (2015) examined two outcomes (i.e., listening and reading for English learning); Li (2018) measured three outcomes (i.e., vocabulary, reading comprehension, and morphological awareness for English learning); Yufrizal and Hasan (2017) assessed two outcomes (i.e., English oral performance and English competence for English learning); García-Álvarez de Perea and Ramírez-García (2024) measured two courses (i.e., Introduction to Accounting and Intermediate Financial Accounting for content learning); and Sato and Hemmi (2022) measured two outcomes (i.e., speaking and writing for English learning). For most samples, we manually computed Cohen's *d* using the reported descriptive statistics, such as means, standard deviations, and sample sizes. However, we also used various estimates for effect size calculation, including beta coefficients from regression analyses for Lin and Lei (2021), Arco-Tirado et al. (2018), Lei and Hu (2014), and Civan and Coskun (2016); *t*-test results for Yang (2015), del Campo et al. (2023), and Salamanca and Montoya (2018); and ANCOVA results for Guo et al. (2018).

Table 1
Overview of 41 independent samples from 28 studies.

Study	Title	Sample	N	Content Learning	English Learning	Within-Group	Between-Group
1	Osam & Korun (2016)	1	39	x	o	o	x
2	Joe & Lee (2013)	2	61	o	x	o	x
3	Lin & Lei (2021)	3	194	o	x	x	o
4	Satayev et al. (2022a)[1]	4	41	x	o	o	x
5	Rogier (2012)	5	58	x	o	o	x
6	Coşgun & Hasırcı (2017)	6	104	x	o	o	x

Study	Title	Sample	N	Content Learning	English Learning	Within-Group	Between-Group
7	Aizawa et al. (2023)[1]	7	46	o	x	o	x
	Aizawa et al. (2023)[2]	8	46	o	x	x	o
8	Aguilar and Muñoz (2014)	9	63	x	o	o	x
9	Satayev et al. (2022b)[1]	10	12	o	x	o	x
	Satayev et al. (2022b)[2]	11	12	o	x	o	x
	Satayev et al. (2022b)[3]	12	12	x	o	o	x
	Satayev et al. (2022b)[4]	13	12	x	o	o	x
10	Yang (2015)	14	29	x	o	o	x
11	Li (2018)	15	53	x	o	o	x
12	Arco-Tirado et al. (2018)	16	206	o	x	x	o
13	Yang et al. (2019)[1]	17	256	o	x	x	o
	Yang et al. (2019)[2]	18	247	o	x	x	o
14	del Campo et al. (2016)[1]	19	45	o	x	x	o
	del Campo et al. (2016)[2]	20	66	o	x	x	o
	del Campo et al. (2016)[3]	21	98	o	x	x	o
15	Thi et al. (2023)[1]	22	29	x	o	o	x
	Thi et al. (2023)[2]	23	29	x	o	x	o
	Thi et al. (2023)[3]	24	78	o	x	x	o
16	Lin & He (2019)	25	498	o	x	x	o
17	Guo et al. (2018)[1]	26	39	o	x	o	x
	Guo et al. (2018)[2]	27	39	o	x	x	o
	Guo et al. (2018)[3]	28	39	x	o	o	x
	Guo et al. (2018)[4]	29	39	x	o	x	o
18	del Campo et al. (2023)	30	424	o	x	x	o
19	Lei and Hu (2014)	31	136	x	o	x	o
20	Yufrizal & Hasan (2017)	32	88	x	o	o	x
21	Pellicer-Ortín et al. (2016)	33	53	o	x	x	o
22	Civan & Coskun (2016)	34	4,151	o	x	x	o
23	Badrie & Abir (2018)	35	21	o	x	o	x
24	Salamanca & Montoya (2018)[1]	36	63	x	o	o	x
	Salamanca & Montoya (2018)[2]	37	63	x	o	x	o
25	Bälter et al. (2024)	38	815	o	x	x	o
26	García-Álvarez de Perea and Ramírez-García (2024)	39	122	o	x	x	o
27	Kim & Kim (2022)	40	130	o	x	o	x
28	Sato & Hemmi (2022)	41	191	x	o	o	x

5.4. Coding of moderator variables

The coding results for the moderator variables, addressing the third research question, are shown in Table 2. For content learning in within-group samples, the moderators “L1-English Related,” “Longer than Semester,” and “Entry Prerequisite” lacked values for certain categories, making data analysis for these variables infeasible. The “Academic English” variable, coded exclusively for English learning, was not applicable in this context. In contrast, for content learning in between-group samples, all seven moderators had values for each category, except for the “Academic English” variable. For English learning in within-group samples, all eight moderators had values for each category, allowing for comprehensive analysis. However, for English learning in between-group samples, the moderators “Full EMI,” “Homogeneity Check,” and “Entry Prerequisite” lacked values for certain categories, preventing moderator analysis for these variables. These findings highlight the variability in data availability across target domains and study designs, affecting the feasibility of certain moderator analyses.

5.5. Data analysis 1: computation of overall average effect size

To address the first and second research questions, we included forest plots (Fig. 2–5) and calculated overall average effect sizes for four samples across target domains and study designs. The forest plots display effect sizes at the sample level, with measurement-level effect sizes averaged for clarity. These plots, accompanied by 95 % confidence intervals, provide a visual summary of the overall trends in the effectiveness of EMI in enhancing students’ content and English learning.

For content learning in within-group samples (see Fig. 2), all but one study, Guo et al. (2018) [1], demonstrated significant positive effect sizes. Guo et al. (2018) reported a non-significant average effect size ($d = 0.19$, $p > .05$, 95 % CI [–0.55, 0.93]). Overall, the EMI programmes included in this meta-analysis appear to have a statistically significant positive impact on students’ content learning. The computation of the overall average effect size yielded a significant large effect size ($d = 1.57$, $p < .001$, 95 % CI [0.89, 2.25]). This indicates substantial gains over time in students’ content knowledge in EMI contexts. However, as within-group effect sizes do not establish causality, these gains cannot be directly attributed to EMI, as other factors may have contributed. Nonetheless, they do indicate that across studies, students enrolled in EMI programmes do appear to successfully acquire content knowledge.

For content learning in between-group samples (see Fig. 3), seven samples had negative average effect sizes, while nine samples had positive average effect sizes. However, when considering their 95 % confidence intervals, only five samples demonstrated statistically significant results. Among these, two had negative average effect sizes, and three had positive

average effect sizes. Overall, the EMI programmes included in this meta-analysis appear to have a comparable pedagogical impact on students' content learning when compared to non-EMI programmes. The computed overall average effect size was $d = 0.13$ ($p > .05$, 95 % CI [–0.04, 0.31]), indicating a non-significant marginal effect size. This suggests that, despite some variability across studies, EMI on the whole does not harm students' content learning and may allow them to perform at levels comparable to those studying in their L1.

For English learning in within-group samples (see Fig. 4), all but three studies—Guo et al. (2018)[3], Aguilar and Muñoz (2014), and Satayev et al. (2022b)[4]—demonstrated significant positive effect sizes, while these remaining three showed non-significant average effect sizes. Overall, the EMI programmes included in this meta-analysis appear to have a statistically significant positive impact on students' English learning. The computation of the overall average effect size yielded a significantly large effect size ($d = 0.81$, $p < .001$, 95 % CI [0.50, 1.12]). This suggests meaningful improvements in students' English proficiency over time in EMI contexts. However, within-group designs lack causal inference, and these observed improvements may be influenced by external factors, such as increased English exposure outside of EMI instruction. Such factors are outside the remit of this meta-analysis.

For English learning in between-group samples (see Fig. 5), all four samples demonstrated positive average effect sizes. However, three of these were not statistically significant, as their 95 % confidence intervals include zero. Overall, the EMI programmes analyzed in this meta-analysis appear to have a modest pedagogical advantage in enhancing students' English learning compared to non-EMI programmes. The computed overall average effect size was $d = 0.33$ ($p = .009$, 95 % CI [0.08, 0.57]), representing a significantly small effect size. The limited sample size ($n = 4$, $k = 4$) weakens the reliability of this estimate, making it unsuitable for drawing generalizable conclusions about the impact of EMI on English learning.

Table 2
Number of categorical values for the eight moderator variables.

Moderator	**Content Learning**				**English Learning**			
	Within-Group ($n = 7$, $k = 7$)		**Between-Group ($n = 19$, $k = 16$)**		**Within-Group ($n = 21$, $k = 14$)**		**Between-Group ($n = 4$, $k = 4$)**	
	True	**False**	**True**	**False**	**True**	**False**	**True**	**False**
L1-English Related	0	7	8	8	2	12	1	3
Institutional Support	1	6	5	11	4	10	2	2
Full EMI	3	4	8	8	6	8	0	4
Longer than Semester	0	7	8	8	3	11	1	3
STEM Course	4	3	6	10	6	8	2	2
Academic English	NA	NA	NA	NA	2	12	2	2
Homogeneity Check	2	5	12	4	3	11	4	0

Moderator	Content Learning				English Learning			
	Within-Group (n = 7, k = 7)		Between-Group (n = 19, k = 16)		Within-Group (n = 21, k = 14)		Between-Group (n = 4, k = 4)	
	True	False	True	False	True	False	True	False
Entry Prerequisite	0	7	2	14	1	13	0	4

Note. The number of categorical values for the moderator variables is determined at the sample level.

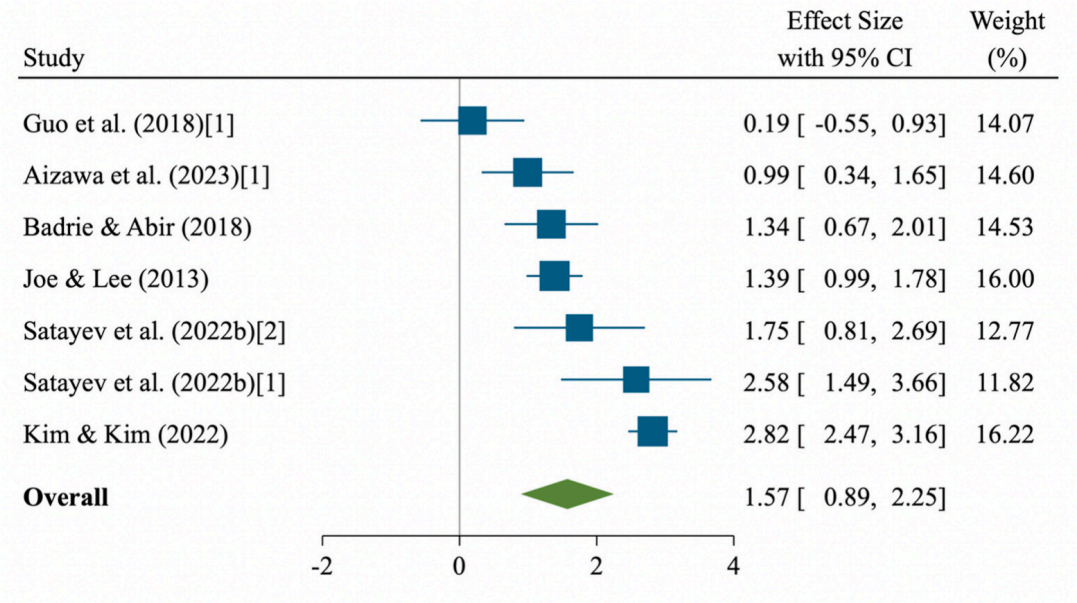


Fig. 2. Forest plot of effect sizes at the sample level for content learning in within-group samples.

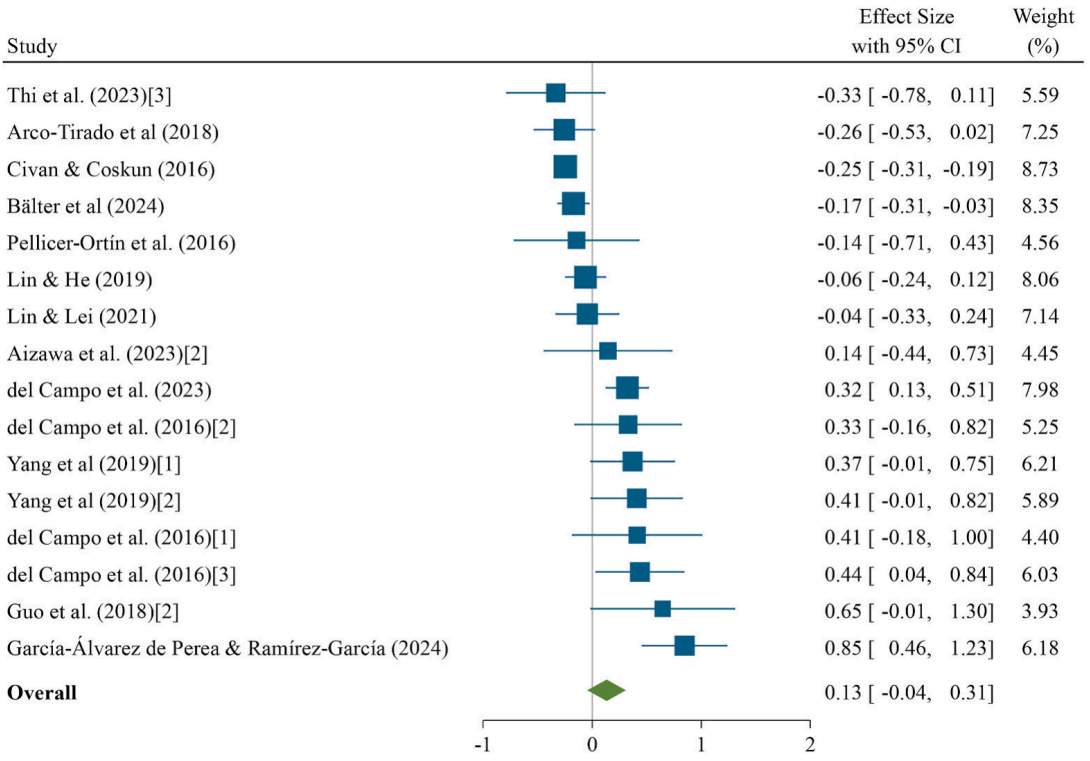


Fig. 3. Forest plot of effect sizes at the sample level for content learning in between-group samples.

5.6. Data analysis 2: analysis of moderation effects

Table 3 presents the results of the analyses of moderation effects. For content learning, the analysis revealed that “Full EMI” had significant moderator effects in within-group samples, while “Homogeneity Check” demonstrated significant moderation effects in both within-group and between-group samples. However, no significant effects were observed for the other moderators in either within-group or between-group samples. In contrast, for English learning, none of the moderators exhibited significant moderation effects in either within-group or between-group samples.

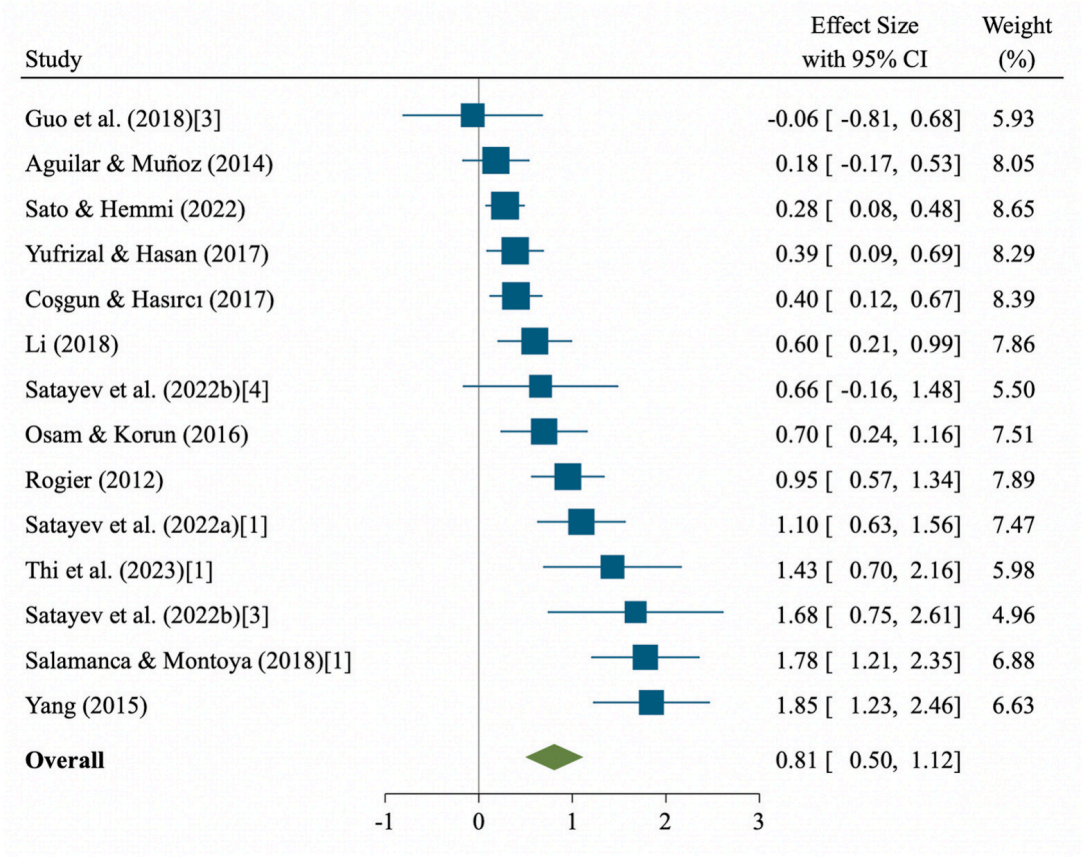


Fig. 4. Forest plot of effect sizes at the sample level for English learning in within-group samples.

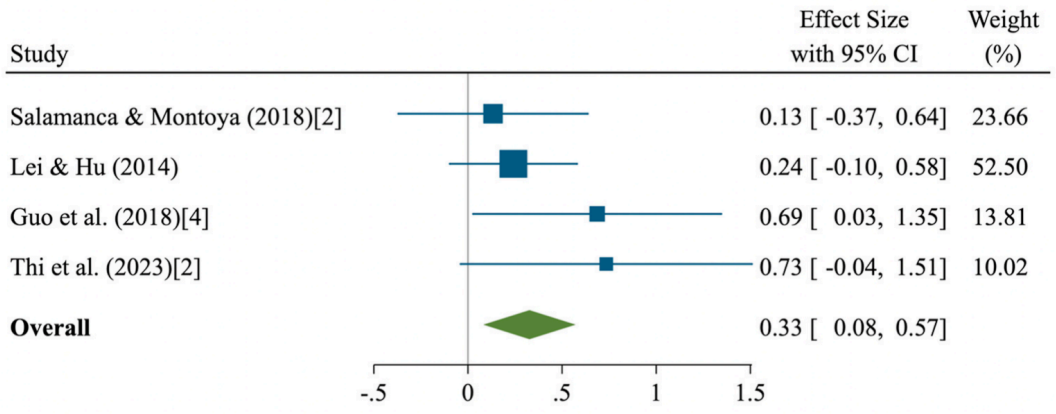


Fig. 5. Forest plot of effect sizes at the sample level for English learning in between-group samples.

First, as shown in Fig. 6, for content learning in within-group samples, we found a subset of studies from full EMI contexts had a significantly larger overall impact (d = 2.48, p < .001, 95 % CI [1.83, 3.13]) than another subset of studies from partial EMI contexts (d = 1.03, p < .001, 95 % CI [0.52, 1.54]).

Second, as shown in Fig. 7, for content learning in within-group samples, we found that studies that did not control for potential selection biases—such as the possibility that more talented and motivated students self-select into EMI programmes—reported inflated effect sizes (d = 1.96, p < .001, 95 % CI [1.31, 2.61]), further overestimating EMI's impact.

Lastly, as illustrated in Fig. 8, our analysis of content learning in between-group samples revealed that studies not accounting for homogeneity (d = 0.41, p < 0.05, 95 % CI [0.04, 0.77]) indicated an apparent advantage of EMI. However, this may reflect pre-existing differences rather than the instructional method itself. In contrast, studies that accounted for homogeneity reported a smaller average effect size (d = 0.04, p > 0.05, 95 % CI [−0.13, 0.20]), underscoring the importance of cautious interpretation.

Table 3
Results of moderator analysis.

Moderator	Content Learning				English Learning			
	Within-Group (n = 7, k = 7)		Between-Group (n = 19, k = 16)		Within-Group (n = 21, k = 14)		Between-Group (n = 4, k = 4)	
	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE
L1-English Related	–	–	0.14	0.18	0.13	0.46	−0.30	0.36
Institutional Support	−0.22	1.04	0.21	0.19	0.44	0.33	−0.50	0.29
Full EMI	1.46**	0.42	0.11	0.18	0.10	0.33	–	–
Longer than Semester	–	–	0.11	0.18	0.26	0.39	−0.21	0.34
STEM Course	0.13	0.76	−0.14	0.19	0.53	0.30	−0.04	0.35
Academic English	NA	NA	NA	NA	−0.14	0.50	0.50	0.29

Moderator	Content Learning				English Learning			
	Within-Group (*n* = 7, *k* = 7)		Between-Group (*n* = 19, *k* = 16)		Within-Group (*n* = 21, *k* = 14)		Between-Group (*n* = 4, *k* = 4)	
	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE
Homogeneity Check	−1.36*	0.60	−0.38*	0.19	0.35	0.40	–	–
Entry Prerequisite	–	–	0.29	0.27	0.31	0.62	–	–

Note. Each row is a moderator variable — a factor that might influence how effective EMI is. Each column group (e.g., "Content Learning – Within-Group") shows the results of testing whether that moderator had a statistically significant impact on effect sizes in that subgroup.
Coeff. = the size of the moderation effect (i.e., how much the effect size changes depending on that moderator).
SE = standard error (reflects the uncertainty of the estimate).
**$p < .01$, *$p < .05$.

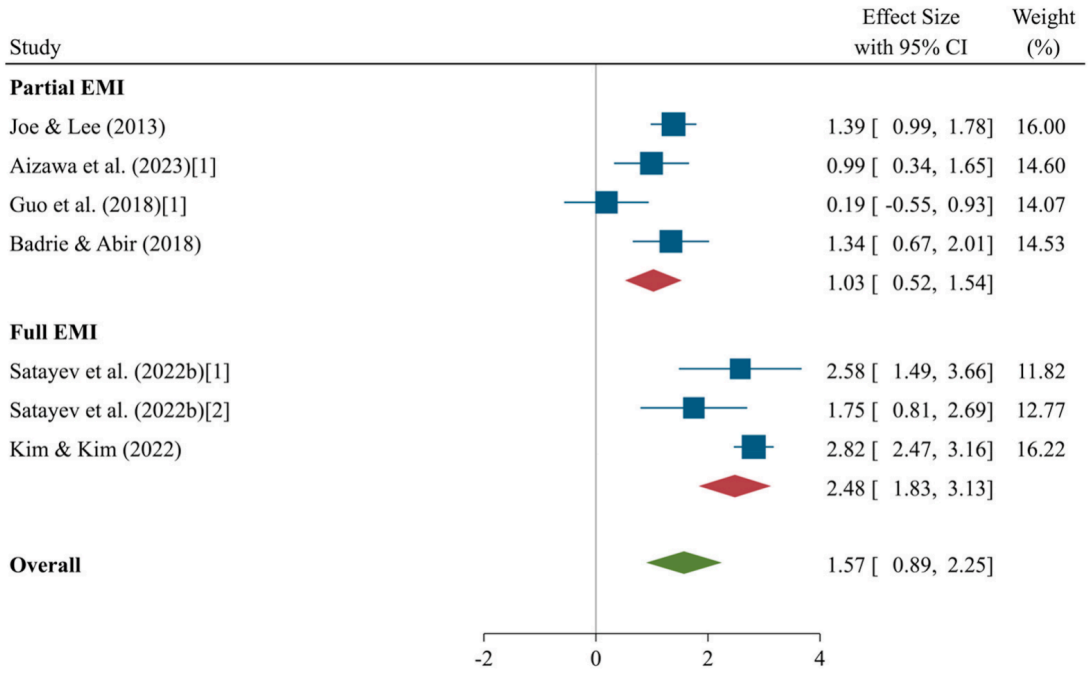


Fig. 6. Ful EMI vs. Partial EMI for Content Learning in Within-Group Samples.

6. Discussion

6.1. Achieving dual goals of English Medium Instruction in higher education

While causal conclusions cannot be definitively drawn from within-group samples, the evidence across a range of studies suggests that EMI successfully fulfills its dual goals: primarily enhancing learners' content knowledge and secondarily fostering English skill development (Rose et al., 2019). This is evident in the large overall average effect sizes observed for both content learning ($d = 1.57$, $p < .001$; 95 % CI [0.89, 2.25]) and English learning ($d = 0.81$, $p < .001$, 95 % CI [0.50, 1.12]).

The substantial gains in content knowledge align with Cummins' developmental interdependence and threshold hypotheses (Cummins, 2000, 2021), which emphasize the importance of attaining sufficient proficiency in both L1 and L2 to benefit from bilingual education. Since students in higher education typically possess stronger language proficiency and continue to develop their academic literacy (Guglielmi, 2008; McCarron & Kuperman, 2022), they are more likely to cross the threshold required to access academic content effectively. Furthermore, the large overall effect size is particularly encouraging, given the cognitive and linguistic demands of higher education content (Chin & Li, 2021). On the whole, the samples in this meta-analysis do not indicate that EMI exerts a cognitive load on learners that negatively impacts their overall content knowledge and linguistic development. However, this finding should be interpreted cautiously, as methodological limitations inherent in EMI studies—discussed in detail in the *Methodological Considerations for English Medium Instruction in Higher Education* section—may impact the reliability of these results.

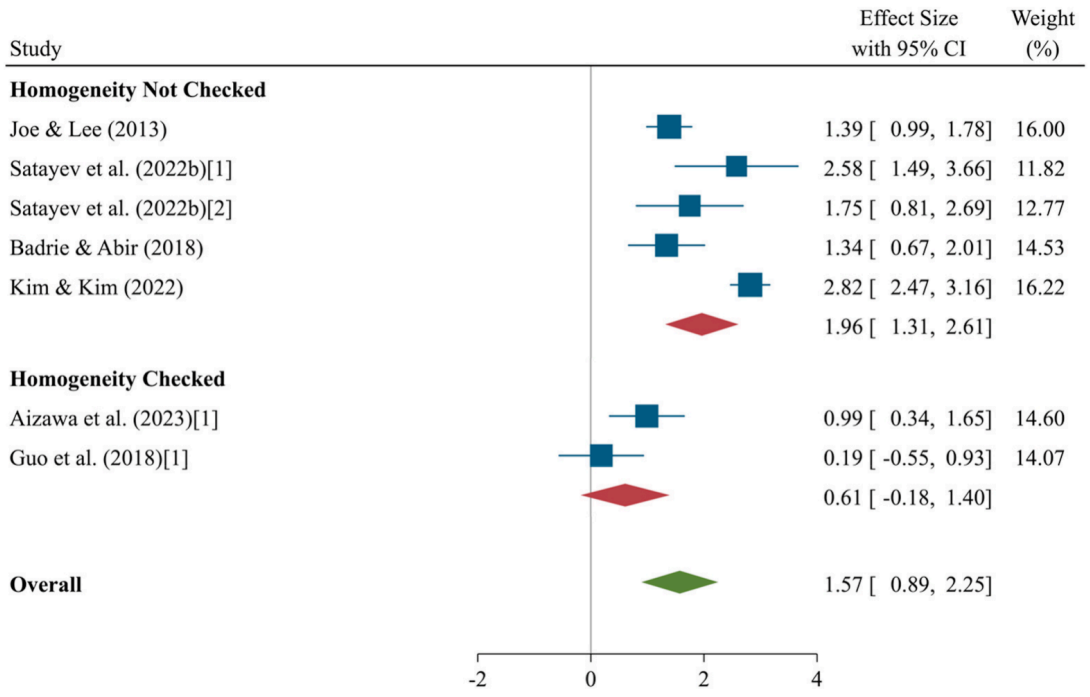


Fig. 7. Homogeneity Checked vs. Unchecked for Content Learning in Within-Group Samples.

With regard to English proficiency development, several factors likely contribute to the positive outcomes. The increased exposure to English input in EMI programmes and greater opportunities for authentic interaction in English are key elements (e.g., J.H. Lee et al., 2023; Martínez Agudo, 2022; Muñoz, 2015). Unlike traditional EFL lessons, the input and interaction in EMI contexts are more closely tied to real-world communication, as students use English as a tool for learning subject content (Gallardo-del-Puerto et al., 2020). This authenticity in language use may further enhance students' English proficiency development.

6.2. Low costs concerning content learning and benefits for enhancing English proficiency

A critical consideration in evaluating the effectiveness of EMI is balance between gains in English proficiency and potential trade-offs in content learning outcomes (Macaro, 2018). This evaluation hinges on two key propositions: (1) EMI should not negatively affect content learning compared to non-EMI instruction where academic subjects are taught in English, consistent with a cost-benefit perspective (Macaro, 2022); and (2) EMI will lead to superior English learning outcomes compared to its non-EMI counterpart. Addressing these propositions requires studies involving between-group comparisons (Cohen et al., 2018), which include non-EMI groups as controls.

The first proposition—that EMI should not detrimentally impact content learning—was supported, as no significant negative effect on academic content learning was observed. This finding is somewhat unexpected, in light of cognitive load theory (Curle et al., 2024), which posits that an excessive cognitive load is anticipated in such an instructional context, especially for students with insufficient English proficiency. A possible explanation for this unexpected finding could be accounted for by Cummins' developmental interdependence hypothesis and the threshold hypothesis. That is, EMI students in higher education are likely to have passed the proficiency threshold in both L1 and L2, and therefore, would not be disadvantaged in terms of content learning, compared to those in non-EMI contexts. In fact, these students appear to perform at comparable levels (d = 0.13, p = .14, 95 % CI [–0.04, 0.31]). This finding aligns with a recent meta-analysis on EMI-CLIL effects on secondary students' content learning, which similarly reported no significant differences between EMI-CLIL and mainstream groups (Kaiypova et al., 2025). However, this finding does not suggest that the EMI and non-EMI conditions are identical for students in terms of their learning experience. Previous research has shown that EMI students often must work harder in order to succeed in learning content (Aizawa et al., 2023). Other studies show EMI peers rely on the use of adaptive learning strategies, particularly listening-related strategies, to process English input effectively (e.g., Zhou & Thompson, 2023). Additionally, intrinsic motivation and self-efficacy may enhance students' ability to deeply engage with and process content, enabling them to match the content knowledge acquisition of non-EMI groups (Zhou et al., 2023).

The second proposition—that EMI improves English proficiency compared to L1 instruction—was also supported, though the overall average effect size was small (d = 0.33, p = .009, 95 % CI [0.08, 0.57]) and derived from a limited sample (n = 4, k = 4). This positive outcome likely stems from the extensive amount of specialised English input present in EMI programmes (e.g., J.H. Lee et al., 2023; Martínez Agudo, 2022; Muñoz, 2015), which is a

notion supported in previous EMI research (Lo & Murphy, 2010). However, the small number of between-group studies focusing on English learning (k = 4) limits the reliability of this conclusion. Further research with larger samples is needed to confirm these findings.

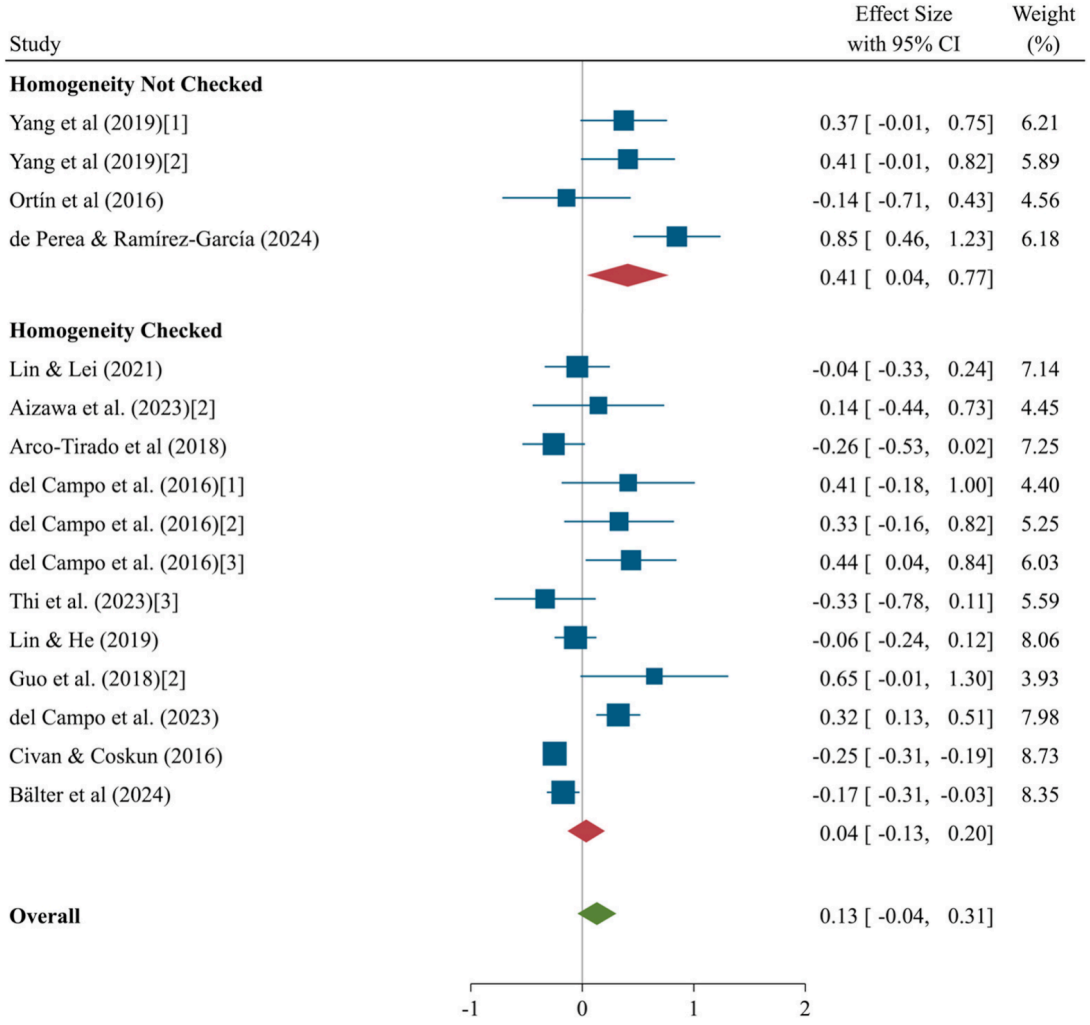


Fig. 8. Homogeneity Checked vs. Not Checked for Content Learning in Between-Group Samples.

6.3. Pedagogical benefits of 'full' English Medium Instruction in higher education

One of the significant moderators identified was the distinction between full and partial EMI contexts for content learning outcomes. From within-group samples, studies conducted in full EMI contexts demonstrated a significantly larger overall impact on learners' content learning (d = 2.48, p < .001, 95 % CI [1.83, 3.13]) compared to those in partial EMI contexts (d = 1.03, p < .001, 95 % CI [0.52, 1.54]). A possible explanation for this result lies in the enhanced opportunities for students in full EMI contexts to acquire both general academic English vocabulary (Coxhead, 2024) and subject-specific vocabulary, along with improved communicative competence (Jablonkai, 2021). These linguistic gains likely enable students to better comprehend academic instruction and, consequently, grasp content knowledge more effectively.

These findings, however, may obscure the reality that in many full EMI programmes, in-sessional, pre-sessional, or preparatory year language training may be offered to students (Macaro, 2018), bolstering their ability to succeed in content learning. This highlights the potential benefits of equipping prospective EMI students with essential linguistic tools before they enter full EMI programmes. Indeed, previous EMI research on transition to EMI from high school has highlighted the importance of offering tailored pathways for students to enhance their academic literacy, especially if entering with lower baseline general proficiency, lower levels of academic vocabulary, or without prior EMI learning experience (Aizawa & Rose, 2020). For instance, preparatory programmes at EMI institutions could focus on teaching the *Academic Spoken Word List* (Dang et al., 2017), which covers over 90 % of academic speech, alongside discipline-specific vocabulary. Providing these resources may help bridge linguistic gaps and enhance students' ability to thrive in full EMI settings.

6.4. Methodological Considerations for English Medium Instruction in Higher Education

Our findings have identified a significant discrepancy across various studies suggesting a potential overestimation of EMI's impact on content learning. The observed benefits reported by some studies may partly reflect pre-existing differences among students, rather than the instructional method itself. This issue arises in both between-group comparisons (e.g., EMI vs. non-EMI) and within-group analyses (e.g., EMI studies using pre-test and post-test measurements).

First, methodological flaws were evident in between-group studies that fail to assess group homogeneity. Studies that did not account for homogeneity or the possibility of selection bias often reported an apparent advantage for EMI. However, this advantage may primarily reflect pre-existing differences rather than the instructional method itself, underscoring the need for cautious interpretation. The recurring issue of group homogeneity in EMI research (and similar instructional approaches) has been noted extensively in the literature (e.g., Bruton, 2011a, 2011b; Graham et al., 2018; Lo & Lo, 2014). Specifically, it has been suggested that experimental groups in these studies often consist of students with higher motivation and aptitude. This pattern is consistent with findings from prior meta-analyses of secondary-level students (Kaiypova et al., 2025; Lo & Lo, 2014), demonstrating that this methodological limitation is not confined to higher education contexts. When restricting analyses to studies that confirmed comparable baseline characteristics between EMI and non-EMI groups, the results suggest that EMI does not negatively affect content learning (d = 0.04, p = .68, 95 % CI [–0.13, 0.20]). These findings, derived from more controlled samples, provide stronger and more reliable evidence that EMI enables learners to achieve performance levels comparable to instruction in their L1. To ensure group homogeneity, one possible approach is to administer a

comprehensive set of tests and surveys assessing students' English proficiency, content knowledge, and affective dimensions (e.g., anxiety, motivation, grit) to both EMI and comparison groups as baseline measures. Then, researchers could employ propensity score matching, in which participants from the EMI group are matched with similar participants from the non-EMI (i.e., comparison) group based on their propensity scores for the target variables. This method would help simulate randomization and reduce the effects of confounding variables, especially since randomization is generally not feasible in EMI research.

Second, within-group analyses exhibit a similar pattern. Studies that failed to control for potential selection biases, such as the tendency for more talented and motivated students self-select into EMI programmes, reported inflated effect sizes. In contrast, studies that ensured comparable characteristics among EMI participants revealed more moderate and realistic differences between pre-test and post-test outcomes (d = 0.61, p > .05, 95 % CI [–0.18, 1.40]), representing a medium-sized effect. These findings further support the conclusion that EMI's impact on content learning is comparable to L1 instruction when methodological rigor is applied. However, the overall average within-group effect size for content learning, without accounting for selection bias, was notably larger (d = 1.96, p < .001; 95 % CI [1.31, 2.61]). This highlights a risk of uncritically accepting evidence of EMI's effectiveness from studies that embody less rigorous designs.

7. Conclusion

In this study, we conducted a multi-level meta-analysis to synthesise quantitative findings on the effectiveness of EMI in enhancing college students' content and English learning. Our results indicate that EMI improves English learning and has a comparable impact on content learning relative to instruction in students' L1. These findings are supported by the methodological rigor of our meta-analytic approach, which incorporates both within- and between-group designs (see Appendix 2 for additional methodological details of the studies included in the meta-analysis). Notably, the comparable impact of EMI on content learning holds even when controlling for whether studies ensured homogeneity between EMI and non-EMI groups—a key concern in prior synthetic reviews of EMI at other educational levels (e.g., Graham et al., 2018; Kaiypova et al., 2025; J.H. Lee et al., 2023; Lo & Lo, 2014). These results further validate the view that EMI can serve as an effective pedagogical approach (Macaro et al., 2018). However, the non-comparability of EMI and non-EMI groups in terms of learner characteristics remains a significant threat to the internal validity of findings from between-subject samples across educational contexts. To address this limitation, we call for more rigorously designed empirical studies on EMI. Future research should prioritize ensuring

comparability between EMI and non-EMI groups through advanced methodological approaches. For within-group designs, researchers should incorporate strategies to control for selection bias, such as accounting for the possibility that more talented, more motivated, or more advantaged (through social class) students may self-select into EMI programmes. Between-group comparisons should, in addition, identify differing levels of content success according to other factors such as academic discipline and proximity of the students' L1 to English.

Some limitations of the present meta-analysis warrant attention. First, datasets for content learning and English learning were derived from different samples, as most studies assessed only one domain; only three studies evaluated both. Similarly, only four studies included both within-group and between-group designs. If more empirical studies include both content learning and English learning outcomes, as well as within-group and between-group designs, future meta-analyses could produce overall effect sizes with lower standard errors, leading to greater precision. Second, the limited number of samples constrained our ability to calculate overall average effect sizes and conduct moderator analyses effectively. For example, the small number of effect sizes for content learning in within-group designs and English learning in between-group designs may not adequately represent broader trends among learners or the complexities of EMI implementation. Third, due to the limited sample size, some moderator variables lacked sufficient categorical diversity, rendering several moderators ineligible for analysis. Even when moderator analyses were conducted, inferential statistics based on only two or three effect sizes were unlikely to yield findings that could be considered generalizable. Therefore, we encourage researchers to provide detailed descriptions of their EMI programmes and to conduct more empirical research examining the effectiveness of EMI to validate our findings.

Overall, this study provides valuable insights into the impact of EMI on content and English learning in higher education while identifying critical areas for future research. We encourage EMI researchers to address these gaps by employing improved research designs and larger, more diverse samples. For example, given the significant evidence of potential selection bias among participants in some EMI programmes, future studies may adopt more sophisticated (yet ethical) methodological approaches, such as random sampling (e.g., randomly selecting participants to ensure each individual has an equal probability of selection) or propensity score matching (e.g., pairing EMI and non-EMI students with similar characteristics across groups to reduce selection bias) to mitigate selection bias and enhance the validity of findings. Such efforts will refine our understanding of EMI's effectiveness across educational contexts and contribute to the development of more effective and pedagogically sound EMI practices.

CRediT authorship contribution statement

Hansol Lee: Writing – original draft, Software, Methodology, Funding acquisition, Data curation. **Heath Rose:** Writing – original draft, Methodology. **Ernesto Macaro:** Writing – original draft, Methodology. **Jang Ho Lee:** Writing – review & editing, Writing – original draft, Methodology, Data curation.

Data statement

Data will be made available on request.

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Declaration of interest

We have nothing to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.system.2025.103755.

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